



Universidade do Minho

Licenciatura em Engenharia Informática
Mestrado Integrado em Engenharia Informática
Inteligência Artificial

Theoretical-Practical Worksheet n.º 3

Topic: Problem Solving

Exercise 1

Consider the following scenario, in which a passenger wants to find a route on the Lisbon subway map. Consider the scenario in which a passenger wants to find the shortest route from the Aeroporto station to Lumiar station in terms of expected “time” cost.



Formulate the game as a demand problem, indicating:

- Type of problem;
- Initial state;
- Final state or target condition;
- State change operators;
- Cost.



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Exercise 2

Consider the following figure and description of a well-known game (Towers of Hanoi).



- In this version of the problem, we present 3 towers (A, B, and C) and 4 discs (D1 to D4);
- Initially, the discs are positioned on tower C, and the goal is to transfer them to tower A;
- In each move, the player can move one disc from one tower to another tower, as long as they do not place that disc on top of a smaller disc.

Formulate the game as a search problem, indicating the type, state representation, initial state, goal state, operators, and cost..